# Epigenetic Age Acceleration in Adult Survivors of Childhood Cancer

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A pre-registered reanalysis of St. Jude Lifetime Cohort DNA methylation data.

## Summary

**Question.** Do adult survivors of childhood cancer show more epigenetic age acceleration than people with no cancer history, and is chest radiotherapy associated with it?

**Design.** Secondary analysis of public Illumina MethylationEPIC data from the St. Jude Lifetime Cohort (NCBI GEO GSE197678). The analysis plan was committed to GitHub before any methylation data were downloaded. Community controls (n = 282) were matched 1:1 on sex and age (±2 years) to adult survivors from SJLIFE1, using demographic data only. Biological age was estimated with PhenoAge (primary), Horvath and Hannum. Age acceleration was the residual of clock age on chronological age.

**Results.** 281 survivors were matched to 282 controls (median age 35.0 years in both; 51% female). Survivors showed **+3.79 years** of PhenoAge acceleration compared with controls (95% CI 2.68 to 4.91; p < 0.001). Horvath (+1.63 years) and Hannum (+2.16 years) agreed. Among survivors, the 68 who had chest radiotherapy showed a further **+4.95 years** (95% CI 3.03 to 6.86; p < 0.001).

**Conclusion.** Adult survivors of childhood cancer have higher epigenetic age relative to chronological age than age- and sex-matched controls, and chest radiotherapy is associated with further epigenetic age acceleration. This reproduces and extends earlier St. Jude findings; the pipeline is open and can be applied to Ugandan survivors.

## Background and aim

Most children treated for cancer in high-income countries now survive into adulthood, but many develop chronic disease early. In the St. Jude Lifetime Cohort (SJLIFE), 98% of survivors assessed had at least one chronic health condition, and an estimated 80% had a serious, disabling or life-threatening condition by age 45 [1]. This pattern resembles accelerated ageing. Radiotherapy and many chemotherapy agents damage DNA in healthy as well as malignant cells, driving cellular senescence, stem cell loss, chronic inflammation and epigenetic change, all hallmarks of ageing [2].

Epigenetic clocks estimate biological age from DNA methylation at selected CpG sites. Horvath's and Hannum's clocks were trained to predict chronological age; Levine's PhenoAge was trained on a phenotypic age built from nine clinical biomarkers and predicts morbidity and mortality [3]. Epigenetic age acceleration (EAA), the gap between clock age and chronological age, has been reported in SJLIFE survivors and linked to radiotherapy, alkylating agents and chronic conditions [4]. Survivors from Africa are almost absent from this research.

**Aim.** To build an open, pre-registered pipeline that tests whether adult survivors of childhood cancer show more EAA than community controls and whether chest radiotherapy is associated with EAA, as a first step towards studying survivors in Uganda.

## Methods

**Pre-registration.** The analysis plan (`analysis_plan.md`) was drafted on 19 October 2026 and committed to the public GitHub repository on 27 October 2026, before any methylation data were downloaded. Eight dated amendments were added before the survivors' methylation data were opened, and a ninth, recording corrections and post hoc checks, after the results were seen (see Reproducibility). No pre-specified analysis was changed.

**Data.** St. Jude Lifetime Cohort DNA methylation data, NCBI GEO SuperSeries GSE197678: community controls with no cancer history (GSE197676, n = 282) and SJLIFE1 survivors (GSE197674, n = 2,138). DNA was extracted from peripheral blood mononuclear cells and profiled on the Illumina MethylationEPIC array. We used the depositors' processed average beta values. Age at sampling, sex and treatment exposures (yes/no) were taken from the GEO sample records. The data are public and de-identified; no attempt was made to identify any participant.

**Selection and matching.** Survivors aged 18 or over at sampling formed the pool (n = 2,068). Each control was matched to one survivor of the same sex with the nearest age, within 2 years, without replacement. Controls were processed in random order and ties broken at random (seed 20261027). Matching used age and sex only, and the list of selected survivors was committed to GitHub before any survivor methylation values were opened. Matching was chosen so that both groups had the same age and sex distribution (the adult survivor pool was younger, median 32.2 vs 35.0 years), and because computing clocks for 281 rather than 2,068 survivors kept the analysis within free cloud computing limits.

**Epigenetic clocks.** Horvath, Hannum and Levine (PhenoAge) clock ages were computed with `DNAmAge()` in the R/Bioconductor package methylclock, without blood cell-count adjustment. Clock CpGs absent from the data were left out of each clock's weighted sum (methylclock does not impute them); missing values at measured CpGs, if any, are

imputed by the package. No further normalisation was applied. Columns in the GEO files were linked to sample records by the sample IDs given in the GEO titles; all 282 control and all 2,138 SJLIFE1 columns were linked.

**Outcome.** For each clock, EAA was the residual from a linear regression of clock age on chronological age, fitted across all 563 participants. A positive EAA means an epigenetic age higher than expected for one's chronological age.

**Statistical analysis.** Primary: linear regression of PhenoAge EAA on group (survivor vs control) and sex. Secondary: the same model for Horvath and Hannum; and, within survivors only, regression of EAA on chest radiotherapy (yes/no) and sex, for all three clocks. Results are differences in years with 95% confidence intervals; significance level 0.05.

**Software and computing.** R with data.table, ggplot2 and methylclock. Clocks were computed in Google Colab (R runtime) in batches of 30 samples; for survivors, only the 281 selected columns were extracted from the GEO file before computing. Package versions are recorded in `results/session_info.txt`.

## Results

**Participants.** Of 2,138 SJLIFE1 survivors, 70 were under 18 at sampling, leaving 2,068 in the pool. 281 were matched to 282 controls; one control, aged 70.2 years, had no survivor within 2 years. Matching was close: the standardised mean difference in age was −0.012, and the mean absolute age gap within pairs was 0.04 years.

**Table 1. Participants**

| Characteristic | Controls (n = 282) | Matched survivors (n = 281) |
|---|---|---|
| Age, mean (SD), years | 35.9 (10.3) | 35.7 (10.1) |
| Age, median (range), years | 35.0 (18.6–70.2) | 35.0 (18.7–66.4) |
| Female, n (%) | 145 (51%) | 144 (51%) |
| Chest radiotherapy, n (%) | – | 68 (24%) |
| Abdominal/pelvic radiotherapy, n (%) | – | 59 (21%) |
| Brain radiotherapy, n (%) | – | 94 (33%) |
| Alkylating agents, n (%) | – | 164 (58%) |
| Anthracyclines, n (%) | – | 146 (52%) |
| Epipodophyllotoxins, n (%) | – | 97 (35%) |

| Characteristic | Controls (n = 282) | Matched survivors (n = 281) |
|---|---|---|
| Platinum agents, n (%) | – | 27 (10%) |
| Corticosteroids, n (%) | – | 137 (49%) |
| Vincristine, n (%) | – | 200 (71%) |

**Clock performance.** All three clocks tracked chronological age closely in both groups (correlation in controls / survivors: Horvath 0.92 / 0.92, Hannum 0.90 / 0.89, PhenoAge 0.88 / 0.88). In controls, Horvath read on average 1.4 years above chronological age, while Hannum and PhenoAge read 10.1 and 19.7 years below it (Figure 1). These constant offsets are removed by the residual definition of EAA.

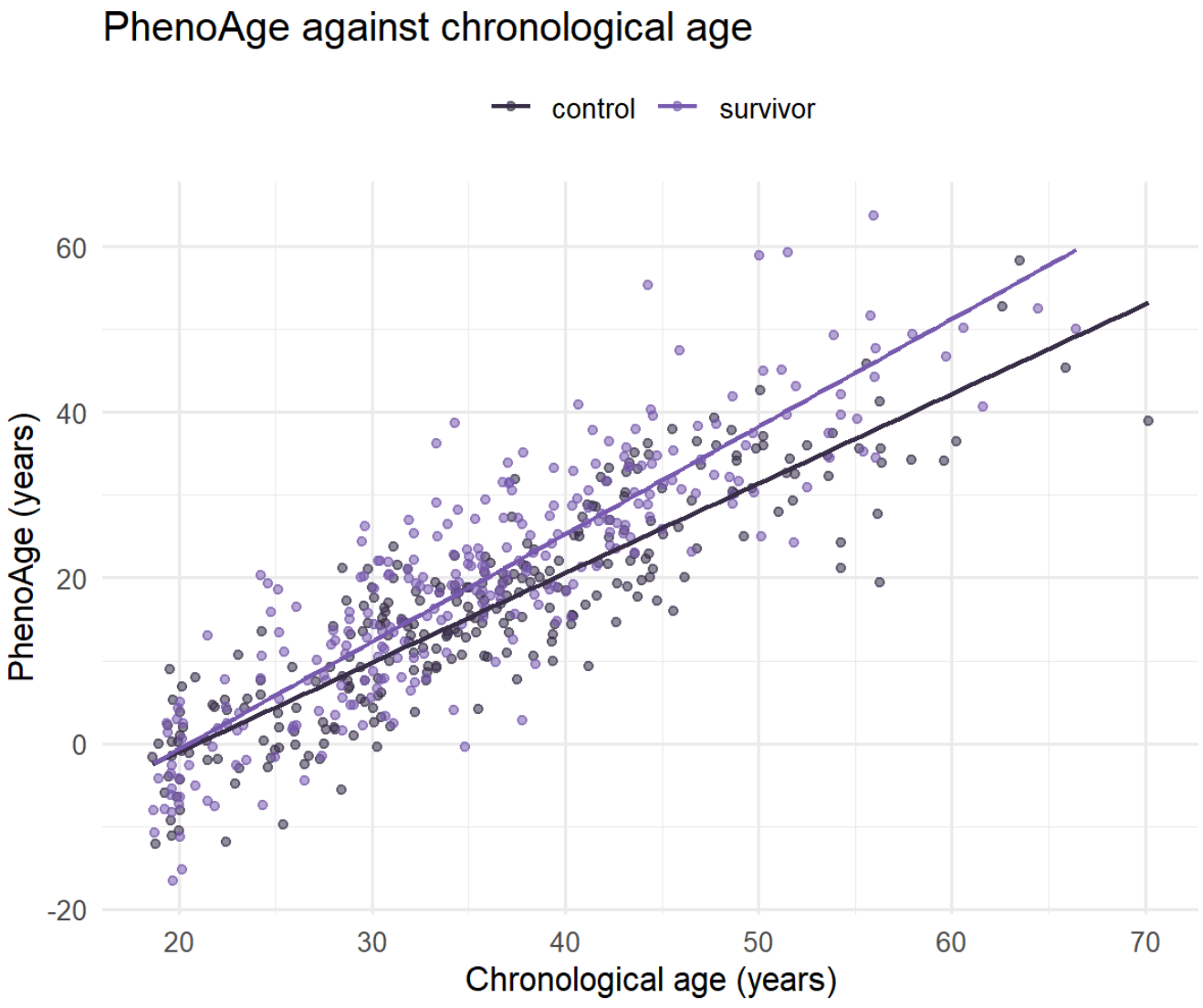



*Figure 1. PhenoAge against chronological age in matched survivors (n = 281) and controls (n = 282), each with a fitted line.*

**Survivors versus controls.** Survivors showed greater age acceleration than controls on all three clocks (Table 2, Figure 2). The primary result was +3.79 years of PhenoAge acceleration (95% CI 2.68 to 4.91; p < 0.001). In a post hoc check, Holm correction across the five secondary tests left all p-values below 0.001.

**Chest radiotherapy.** Among survivors, those who had chest radiotherapy (n = 68) showed more age acceleration than those who did not (n = 213), on all three clocks; by PhenoAge, +4.95 years (95% CI 3.03 to 6.86; p < 0.001) (Table 2, Figure 3).

**Table 2. Differences in epigenetic age acceleration, adjusted for sex**

| Comparison | Clock | n | Difference, years | 95% CI | p |
|---|---|---|---|---|---|
| Survivors vs controls (primary) | PhenoAge | 563 | +3.79 | 2.68 to 4.91 | $5.6 \times 10^{-11}$ |
| Survivors vs controls | Horvath | 563 | +1.63 | 0.97 to 2.29 | $1.5 \times 10^{-6}$ |
| Survivors vs controls | Hannum | 563 | +2.16 | 1.39 to 2.93 | $5.3 \times 10^{-8}$ |
| Chest RT vs none, survivors | PhenoAge | 281 | +4.95 | 3.03 to 6.86 | $6.5 \times 10^{-7}$ |
| Chest RT vs none, survivors | Horvath | 281 | +2.51 | 1.43 to 3.60 | $7.5 \times 10^{-6}$ |
| Chest RT vs none, survivors | Hannum | 281 | +2.72 | 1.39 to 4.05 | $7.5 \times 10^{-5}$ |

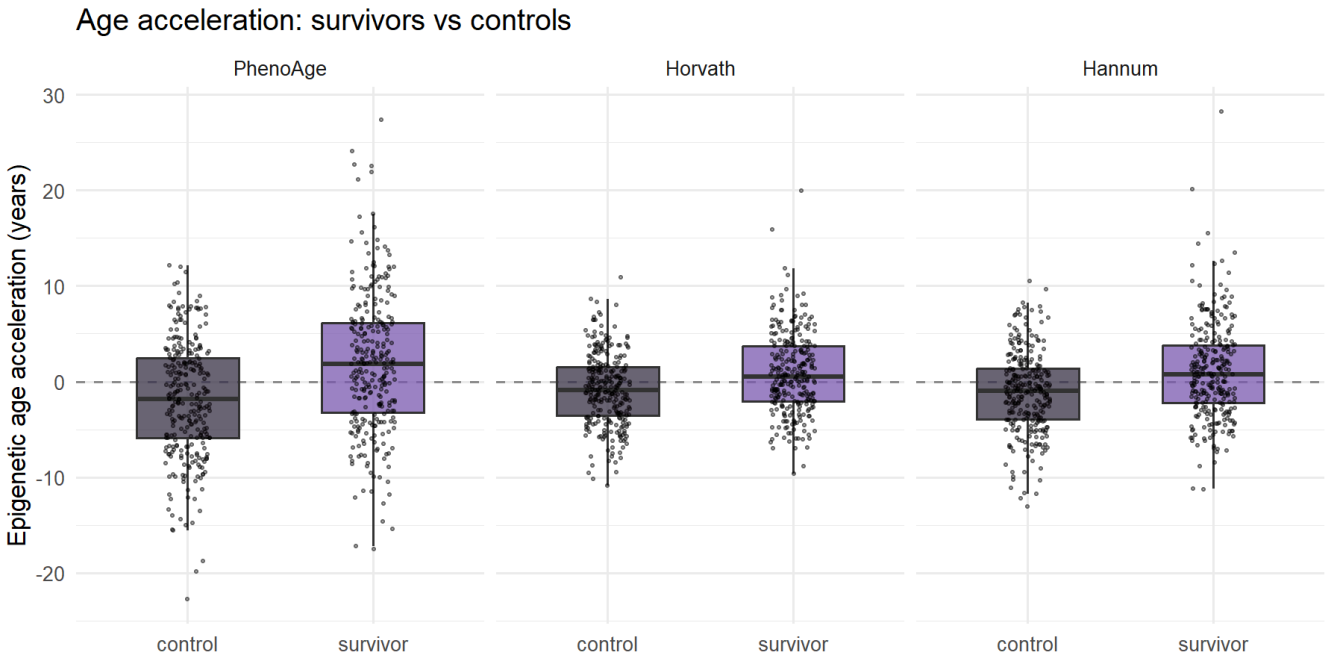



*Figure 2. Epigenetic age acceleration in controls and matched survivors for each clock. Boxes show median and interquartile range; dots are individuals; dashed line at zero.*

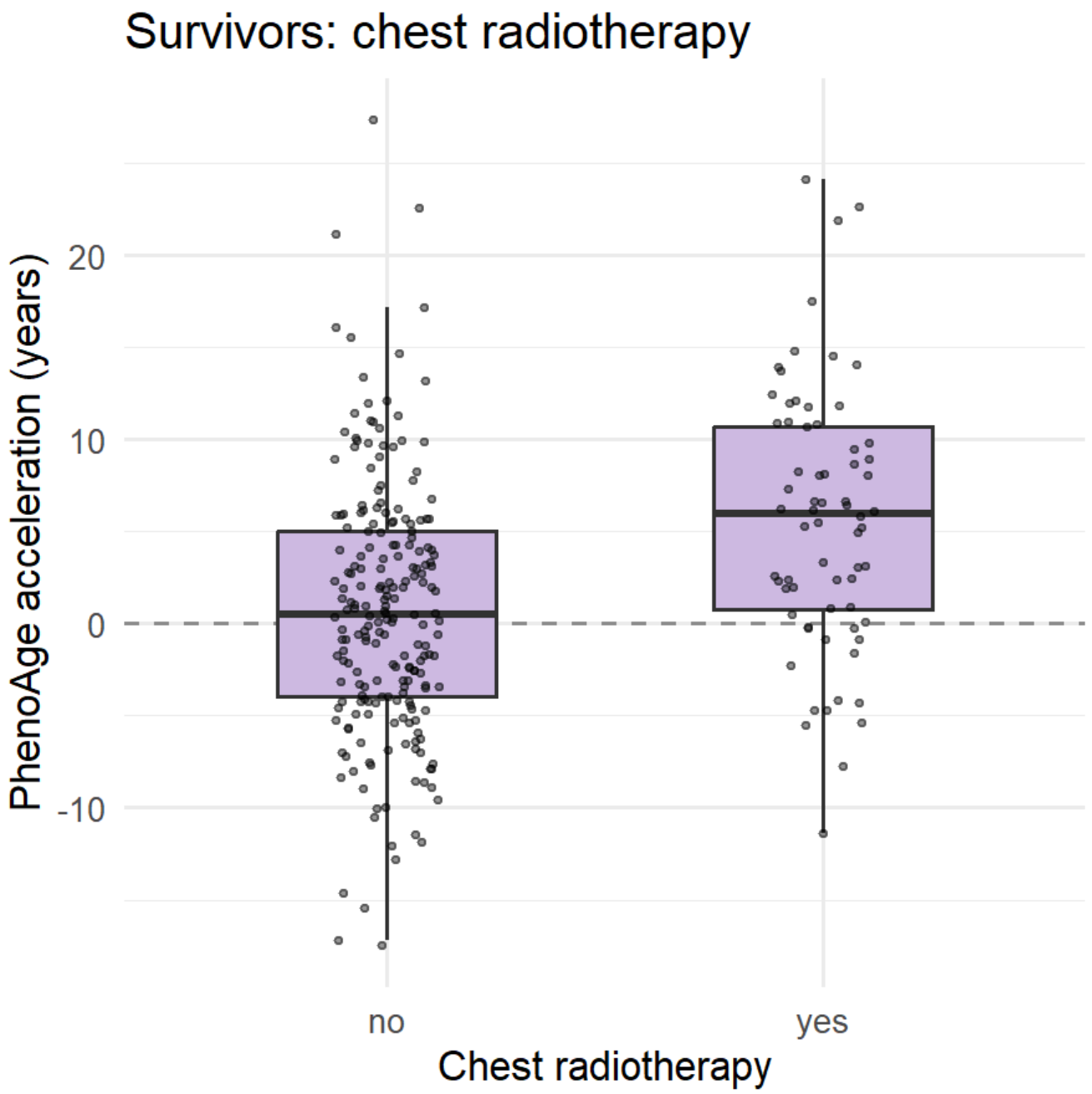



*Figure 3. PhenoAge acceleration in survivors without (n = 213) and with (n = 68) chest radiotherapy.*

**Exploratory analyses (post hoc).** These were added after the primary results were seen, following external review (`scripts/06_sensitivity.R`), and do not replace the pre-specified results.

- *Robustness of the main estimate.* Estimating the group difference directly (clock age ~ age + group + sex) or with the age line fitted in controls only gave the same result: PhenoAge +3.79 and +3.78 years. In standard-deviation units of control age acceleration, the differences were 0.62 (PhenoAge), 0.43 (Horvath) and 0.51 (Hannum).
- *Power.* With 281 per group, the smallest difference detectable with 80% power at alpha 0.05 was 1.44 years for PhenoAge (0.24 SD), well below the observed 3.79 years.

**Table 3. Exploratory models (years, 95% CI)**

| Model | PhenoAge | Horvath | Hannum |
|---|---|---|---|
| Group × age: extra years of clock age per year of chronological age in survivors (n = 563) | +0.22 (0.11 to 0.33), p < 0.001 | +0.08 (0.02 to 0.15), p = 0.012 | +0.15 (0.07 to 0.22), p < 0.001 |
| Chest RT vs none, adjusted for age, sex and eight other treatment flags (n = 281) | +4.21 (0.99 to 7.42), p = 0.011 | +1.42 (−0.48 to 3.32), p = 0.14 | +2.37 (0.05 to 4.69), p = 0.045 |

## Discussion

In this pre-registered reanalysis, adult survivors of childhood cancer had higher epigenetic age relative to chronological age than age- and sex-matched community controls on all three clocks, by nearly four years on PhenoAge. Survivors who had chest radiotherapy showed about five years more PhenoAge acceleration again.

**Pattern across clocks.** The difference was largest for PhenoAge, smallest for Horvath and intermediate for Hannum. This fits what each clock was built to measure. PhenoAge was trained on clinical markers of ill health and mortality, which survivors accumulate early. Horvath's clock was trained to predict chronological age across tissues and is relatively insensitive to blood cell composition. Hannum's blood-based clock lies between the two.

**Comparison with earlier work.** Qin et al. analysed 2,139 SJLIFE survivors against the same 282 controls with PhenoAge and reported adjusted mean EAA of +0.63 years in survivors and −3.61 years in controls, a gap of about 4.2 years; chest radiotherapy, abdominal/pelvic radiotherapy, alkylating agents, epipodophyllotoxins and glucocorticoids were independently associated with EAA [4]. Our estimate of +3.79 years and our chest radiotherapy finding are consistent with theirs. Because this analysis used the same controls and a subset of the same survivors, it is a reproduction with a different design (pre-registered, individually age- and sex-matched, three clocks), not an independent replication.

**Biological interpretation.** Ionising radiation and cytotoxic chemotherapy damage DNA in normal tissue, inducing cellular senescence, stem cell depletion and persistent inflammation, which are hallmarks of ageing [2]. Chest radiotherapy exposes the heart, lungs and the marrow of the sternum, ribs and thoracic spine, which supplies the blood cells measured here. Chest radiotherapy is also often given with alkylating agents, for example for Hodgkin lymphoma, so part of its association may reflect co-treatment.

**Exploratory findings.** The gap between survivors and controls was larger at older ages: by PhenoAge, survivors' clock age rose about 0.22 years more per year of chronological age, so the estimated gap at age 50 is about 6.5 years larger than at age 20. This matches the direction of Qin et al., who reported 1.24 versus 1.08 years of epigenetic ageing per calendar year [4]. Because the data are cross-sectional, this may reflect when older survivors were treated, as treatment has changed over the decades, rather than a faster rate of ageing within each person. After adjustment for age, sex and the other eight treatment flags, chest radiotherapy remained associated with PhenoAge acceleration (+4.21 years), although with a much wider confidence interval; for Horvath the adjusted estimate could not be distinguished from zero. Part of the unadjusted chest radiotherapy association may therefore reflect co-treatment.

**Limitations.**

- Group is confounded with GEO series: controls (GSE197676) and survivors (GSE197674) may have been processed in different batches, and this design cannot separate a batch effect from a true difference. Cross-sectional: one blood sample per person shows a difference in epigenetic age, not a measured rate of ageing.
- The chest radiotherapy model was adjusted for sex only, as pre-specified; an exploratory model adjusted for the other treatments gave a smaller, less precise estimate (Table 3), and neither model accounts for diagnosis or time since treatment, so the estimate is an association.
- No adjustment for blood cell composition. Treatment can shift the cell mix, and clocks partly read that shift.
- Samples were peripheral blood mononuclear cells, while the clocks were developed in whole blood; this explains the large constant offsets in Hannum and PhenoAge, which the residual removes, but cell-type effects may remain.
- Treatment was recorded as yes/no, with no dose or field information; cancer diagnosis is not available in GEO.
- One US cohort. Results may not transfer to African survivors, whose treatments, infections, nutrition and genetic backgrounds differ.

## Conclusion and next steps

Adult survivors of childhood cancer have higher epigenetic age relative to chronological age than matched people with no cancer history, and chest radiotherapy is associated with further epigenetic age acceleration. A pre-registered, fully scripted pipeline built on free tools reproduced this from public data.

Planned next steps, each to be reported as exploratory because they were not in the primary plan:

1. The elastic-net age predictor trained on the controls, compared with Horvath's clock (listed as exploratory in the plan).
2. Age acceleration adjusted for estimated blood cell composition.
3. The comparison repeated in all 2,068 adult survivors with age adjustment, for more power.
4. In the longer term, the same pipeline applied to Ugandan survivors of childhood cancer, which would need new samples, ethical approval and funding.

## Reproducibility

**Code and outputs.** All scripts, the analysis plan with its commit history, summary results and figures are at github.com/Mrn-selyn-commits/sjlife-epigenetic-age. Running scripts 00 to 06 in order reproduces every number in this report. Sample-level data are not stored in the repository; they are downloaded from GEO by the scripts.

**Plan amendments.** Amendments 1 to 8 were made before survivors' methylation data were opened; amendment 9 after the primary results were seen.

| Date | Amendment |
|---|---|
| 27 Oct 2026 | 1. Matching rule made exact: 1:1 within sex, nearest age, caliper 2 years, seed 20261027, survivors aged 18 or over |
| 27 Oct 2026 | 2. Cancer diagnosis is not in GEO and was dropped from the data dictionary |
| 27 Oct 2026 | 3. GEO describes the samples as PBMCs; the publication says blood-derived DNA. Reported as a limitation |
| 27 Oct 2026 | 4. Methylation values are in each series' supplementary processed file, not the series matrix |
| 3 Nov 2026 | 5. Constant clock offsets in controls (Hannum and PhenoAge below age); removed by the residual definition |
| 3 Nov 2026 | 6. GEO labels control samples as "survivor"; treated as a labelling error |
| 3 Nov 2026 | 7. Batching and column extraction are implementation only and do not change estimates |

| Date | Amendment |
|---|---|
| 3 Nov 2026 | 8. Chest radiotherapy comparison reported for all three clocks, PhenoAge first |
| 3 Nov 2026 | 9. After results: wording and reference corrections; how methylclock handles missing CpGs; post hoc checks in script 06; batch confounding added as a limitation |

**Use of AI assistance.** AI tools (Claude, Anthropic) were used responsibly in developing this project.

**References**


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